Empirical Recovery of Distance-Deterrence Parameters from Aggregate Travel-Distance Distributions

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Abstract

Accurate origin–destination (OD) flow estimation is fundamental to transportation planning, accessibility analysis, infrastructure assessment, and urban decision-making. Gravity-model calibration traditionally requires identifying city-specific distance-deterrence parameters from complete observed OD matrices. Aggregate mobility products are increasingly available, but it remains unclear whether aggregate travel-distance distributions are empirically informative enough to support practical parameter recovery without observing OD interactions. This study investigates whether latent distance-deterrence parameters can be recovered directly from aggregate travel-distance distributions within a specified gravity model and aggregate observation framework. Aggregate travel-distance distributions are extracted from Meta Movement Distribution Maps and formulated within a multinomial likelihood framework to recover latent distance-deterrence parameters. The recovered power-law component shows strong agreement with locally calibrated references, while the exponential component remains accurate in absolute error, and the resulting models produce statistically comparable OD prediction performance across the evaluated metropolitan areas without requiring target-city OD observations. The findings provide empirical evidence that, within this setting, aggregate travel-distance distributions are empirically informative enough to support practical empirical recovery of distance-deterrence parameters. Building on this finding, the proposed probabilistic calibration framework supports survey-free gravity-model calibration in data-scarce cities using aggregate mobility observations.

Keywords: Spatial interaction, gravity models, distance deterrence, aggregate mobility data, travel-distance distributions, parameter recovery, survey-free calibration

1 Introduction

Accurate origin–destination (OD) flow estimation is fundamental to transportation planning, accessibility analysis, infrastructure assessment, and urban decision-making (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Spatial interaction modeling represents the core analytical engine for estimating population mobility patterns across urban spaces. Gravity models (Wilson 1971; Tanner 1961) provide one of the most widely used probabilistic formulations of spatial interaction because they explicitly characterize how travel demand emerges from the interplay between origin production, destination attractiveness, and distance deterrence. Their predictive capability, however, depends critically on identifying appropriate city-specific distance-deterrence parameters.

The fundamental bottleneck in spatial interaction modeling lies in observation rather than model formulation. Identifying distance-deterrence parameters traditionally requires observing complete origin–destination flow matrices or individual mobility trajectories. Yet, comprehensive travel surveys are expensive and conducted infrequently (Egu and Bonnel 2020; Hussain et al. 2021), while detailed trajectory records are often inaccessible in practice because of privacy risk and access restrictions (de Montjoye et al. 2013). Consequently, conventional calibration methods fail in data-scarce urban regions worldwide because the necessary origin–destination observations are unavailable.

Recent years have witnessed a rapid growth in publicly available aggregate mobility statistics—including Meta's Movement Distribution Maps (MDM) (Meta 2026) and related open datasets (Oliver et al. 2020; Buckee et al. 2020; Pappalardo et al. 2023)—released by governments, industrial platforms, and large-scale mobility studies. Unlike complete origin–destination matrices, these aggregate statistics remove origin–destination identities while substantially reducing privacy risk and storage requirements. However, they have been used primarily for descriptive analyses and model evaluation rather than as an information source for behavioural inference. This raises a fundamental research question: *Can aggregate travel-distance distributions serve as a sufficient information source for identifying collective mobility behaviour?*

This study is founded on the central empirical hypothesis that *within the specified gravity model and aggregate observation framework, aggregate travel-distance distributions are empirically informative enough to support practical empirical recovery of city-specific distance-deterrence parameters*. To systematically evaluate this hypothesis, we test three sequential empirical expectations. First, aggregate travel-distance observations induce a statistically valid multinomial likelihood for recovering latent distance-deterrence parameters. Second, building on this likelihood, distance-deterrence parameters recovered from aggregate travel-distance distributions show close agreement with reference parameters calibrated from complete origin–destination interactions, with the strongest agreement appearing in the dominant power-law component. Third, building on parameter agreement, distance-deterrence parameters recovered from aggregate observations preserve the downstream predictive performance of gravity-model calibration in unobserved target cities.

The contributions of this study operate through a logical dependency structure across three distinct layers. At the core layer (**Layer 1 — Empirical Finding**), we show that, within the specified gravity model and aggregate observation framework, aggregate travel-distance distributions are empirically informative enough to support practical recovery of city-specific distance-deterrence parameters without observing origin–destination interactions. Building on this result (**Layer 2 — Methodological Contribution**), we establish a probabilistic parameter-identification framework to infer latent distance-deterrence parameters directly from aggregate travel-distance observations and integrate them into gravity-model calibration. Finally (**Layer 3 — Practical Contribution**), we support survey-free gravity-model calibration for data-scarce cities lacking target-city OD observations.

The remainder of this paper is organized as follows. Section 2 reviews related work on spatial interaction modeling, parameter identification, aggregate mobility representations, and survey-free prediction. Section 3 presents the proposed probabilistic parameter-identification framework, establishing its mathematical formulation across five distinct layers. Section 4 empirically evaluates the three scientific expectations across 50 US metropolitan areas. Section 5 discusses scientific implications, scope, limitations, and future directions. Section 6 concludes the paper.

2 Related Work

Gravity models provide one of the most explicit and widely used formulations of spatial interaction, expressing travel demand as the joint effect of origin production, destination attraction, and a distance-deterrence function (Wilson 1971; Fotheringham and O'Kelly 1989; Sen and Smith 1995). Because the deterrence function is represented explicitly, gravity-model calibration has historically focused on estimating city-specific distance-decay parameters rather than on redesigning the model structure itself.

Classical calibration methods assume access to complete observed OD flows. Entropy-based,

Table 1 Representative comparison of calibration paradigms relevant to this study.

Approach	Explicit Deterrence Parameters	Uses Full Aggregate Distribution Shape	Requires Targe Observed Trip
Classical Gravity Calibration (Wilson 1971; Flowerdew and Aitkin 1982)	Yes	No	Yes
Moment-Based Aggregate Calibration (Tanner 1961; Hyman 1969; Merlin 2020)	Yes	No	Yes
DeepGravity (Simini et al. 2021)	No	No	Yes
Proposed Framework (This Study)	Yes	Yes	No

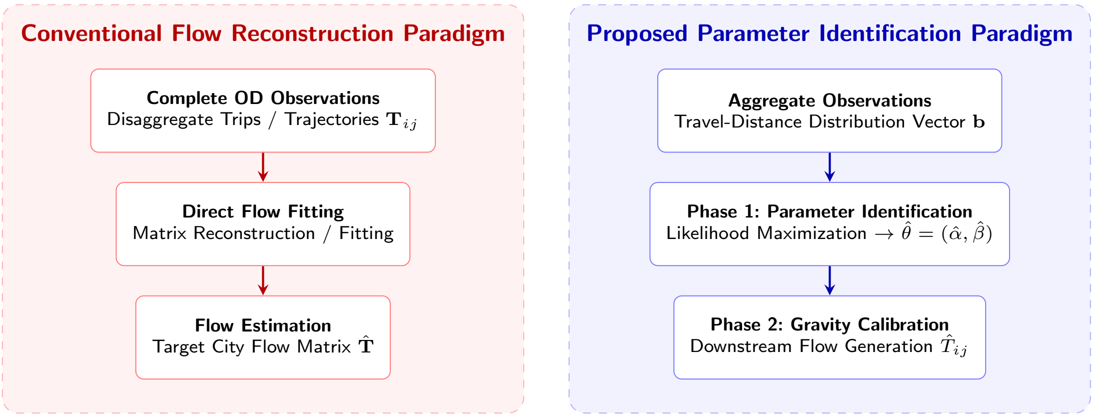

Fig. 1 Conceptual distinction between direct flow reconstruction paradigms and the proposed parameter identification paradigm under aggregate observations. Rather than attempting to reconstruct point-to-point mobility matrices from complete OD surveys, the proposed framework identifies latent distance-deterrence parameters directly from aggregate travel-distance representations prior to downstream spatial interaction generation.

regression-based, and Poisson-likelihood procedures all fit deterrence parameters directly on observed OD matrices (Wilson 1971; Flowerdew and Aitkin 1982; Lenormand et al. 2016). Under this paradigm, the central methodological question has historically been how best to fit the same parametric structure once OD data are available, rather than whether deterrence parameters can be identified when OD interactions are unobserved. For completeness, parameter-free alternatives such as the Radiation model (Simini et al. 2012) avoid explicit deterrence parameters altogether; because this study specifically targets deterrence-parameter recovery, such parameter-free models serve only as a baseline in our experiments rather than as a directly comparable calibration paradigm.

A smaller, classical aggregate-calibration tradition relaxes the requirement of a complete OD matrix, but only partially. Tanner (Tanner 1961) and Hyman (Hyman 1969) calibrate the deterrence parameter by matching the mean observed trip length, while Merlin (Merlin 2020) proposes a median-based alternative for single-parameter spatial interaction models. Yang et al. (Yang et al. 2014) study commuting-flow prediction in the absence of local calibration data, but do not estimate deterrence parameters from a published aggregate travel-distance distribution. These studies establish that aggregate calibration is not new; more fundamentally, each constrains only a single summary statistic of locally observed trips and still requires observed trips from the target city.

Recent deep-learning approaches to spatial interaction, such as DeepGravity (Simini et al. 2021), expand predictive capacity by learning flow-generating functions directly from spatial covariates, and subsequent transfer-oriented variants attempt to reduce target-domain data requirements (Yang et al. 2026; Enaya et al. 2026). These approaches do not close the gap identified above: they are typically trained or fine-tuned using observed OD flows and, unlike the present framework, do not target the explicit deterrence parameters of a gravity specification.

Table 1 summarizes these paradigms along the dimensions most relevant to this study. The unresolved empirical question is therefore narrower than whether aggregate mobility data are useful in general: it is whether the *full shape* of a published aggregate travel-distance distribution, combined with urban structure derived from open data, is empirically informative enough to support practical recovery of distance-deterrence parameters without any observed trips from the target city. This motivates the primary inference-space formulation developed in Section 3, shifting aggregate travel-distance distributions from downstream evaluation targets to primary observation spaces.

3 Probabilistic Parameter-Identification Framework

Rather than attempting to estimate origin–destination (OD) flows directly from aggregate data, we formulate gravity calibration as a probabilistic parameter-identification problem. The primary objective is to identify city-specific distance-deterrence parameters $\theta = (\alpha, \beta) \in \Theta$ from a spatial representation layer $\mathbf{b} = (b_1, \ldots, b_K)^\top \in \Delta^{K-1}$ summarizing population travel-distance distributions. The secondary objective is to evaluate whether the identified parameters $\hat{\theta}$ enable survey-free gravity-model calibration to predict unobserved target-city OD matrices $\hat{\mathbf{T}}$. To achieve this, the proposed framework is structured across five distinct mathematical layers.

Layer 1 — Physical Spatial Interaction Model: We consider a spatial interaction system comprising N_{zones} geographic zones. The expected physical flow T_{ij} from origin zone i to destination zone j under a singly constrained gravity specification is expressed in separable form:

$$T_{ij} = O_i \frac{A_j f(d_{ij}; \theta)}{\sum_m A_m f(d_{im}; \theta)}, \tag{1}$$

where O_i represents origin trip production, A_j represents destination attraction, d_{ij} is inter-zonal travel distance, and $f(d; \theta)$ is a distance-deterrence function parameterized by vector $\theta = (\alpha, \beta)$. Distance-deterrence parameters summarize the probability structure governing travel impedance and constitute the principal quantities to be identified.

Layer 2 — Observation Model: Let $\mathcal{P} : \mathbb{R}^{N_{\text{zones}} \times N_{\text{zones}}} \to \mathbb{R}^K$ denote a linear observation operator projecting origin–destination interactions onto a K-bin aggregate distance distribution:

$$n_k = \mathcal{P}(\mathbf{T})_k = \sum_{i=1}^{N_{\text{zones}}} \sum_{j=1}^{N_{\text{zones}}} T_{ij} \cdot \mathbf{1}(d_{ij} \in \text{bin } k), \quad k \in \{1, \ldots, K\}, \tag{2}$$

where n_k is the empirical trip count in distance bin k, and $N = \sum_{k=1}^{K} n_k$ is total trip volume. The normalized observation vector $\mathbf{b} = (b_1, \ldots, b_K)^\top \in \Delta^{K-1}$ with $b_k = n_k / N$ represents empirical trip-length probabilities across distance intervals k.

Layer 3 — Probabilistic Observation Model: The model-predicted probability $\pi_k(\theta)$ of a trip falling within distance bin k under parameter vector θ is derived by marginalizing joint trip probabilities across zonal pairs:

$$\pi_k(\theta) = \sum_{i,j: d_{ij} \in \text{bin } k} \frac{O_i A_j f(d_{ij}; \theta)}{\sum_m A_m f(d_{im}; \theta)} \approx \frac{E_k f(d_k; \theta)}{\sum_{k'} E_{k'} f(d_{k'}; \theta)}, \tag{3}$$

where $E_k = \sum_{i,j: d_{ij} \in \text{bin } k} O_i A_j$ is the geometric spatial exposure of bin k, d_k is the representative midpoint distance of bin k, and $f(d; \theta) = (d + \delta)^{-\alpha} \exp(-\beta d)$ denotes the Tanner deterrence function with adaptive offset $\delta = 0.5 \bar{R}_{\text{zone}}$. The right-most expression is the bin-midpoint approximation used in implementation to obtain a tractable binned observation model. Assuming N total trips are sampled independently from categorical probability vector $\boldsymbol{\pi}(\theta) = (\pi_1(\theta), \ldots, \pi_K(\theta))^\top$, the aggregate observation count vector $N\mathbf{b}$ follows a multinomial distribution: $N\mathbf{b} \sim \text{Multinomial}(N, \boldsymbol{\pi}(\theta))$ (Casella and Berger 2002; Murphy 2012).

Layer 4 — Likelihood & Parameter Identification: The log-likelihood of observing aggregate distribution $\mathbf{b}$ given parameter vector θ is:

$$\log \mathcal{L}(\theta; \mathbf{b}) = C + N \sum_{k=1}^{K} b_k \log \pi_k(\theta) = C - N \left[H(\mathbf{b}) + D_{\text{KL}}(\mathbf{b} \parallel \boldsymbol{\pi}(\theta \tag{4}$$

where C is a constant, $H(\mathbf{b})$ is empirical entropy, and D_{KL} is Kullback–Leibler divergence (Cover and Thomas 1991). Consequently, Maximum Likelihood Estimation (MLE) is exactly equivalent to cross-entropy minimization (Bishop 2006; Murphy 2012):

$$\hat{\theta} = \arg\min_{\theta\in\Theta} H(\mathbf{b}, \boldsymbol{\pi}(\theta)) = \arg\min_{\theta\in\Theta} \left[-\sum_{k=1}^{K} b_k \log \pi_k(\theta) \right]. \tag{5}$$

Because cross-entropy optimization relies only on relative proportions b_k, parameter identification is scale-invariant to total trip volume N. Origin trip production $\hat{O}_i$ and destination attraction $\hat{A}_j$ are estimated independently using open demographic features and spatial indicators $\hat{A}_j = \frac{1}{2}[\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)]$.

Layer 5 — Downstream Flow Prediction: Once latent deterrence parameters $\hat{\theta} = (\hat{\alpha}, \hat{\beta})$ are identified via Eq. (5), downstream spatial interaction flows $\hat{T}_{ij}$ are generated by combining the identified parameters with independent production and attraction estimates:

$$\hat{T}_{ij} = \hat{O}_i \frac{\hat{A}_j f(d_{ij}; \hat{\theta})}{\sum_k \hat{A}_k f(d_{ik}; \hat{\theta})}. \tag{6}$$

Algorithm 1 summarizes the parameter identification and downstream interaction workflow.

The framework has four practical features in the present study setting: (1) **Feature 1 (Recoverability under the assumed observation model):** Latent distance-deterrence parameters can be recovered empirically under the assumed observation model in the evaluated settings; (2) **Feature 2 (Observation Efficiency):** Complete origin–destination flow observations are unnecessary for recovering city-specific deterrence parameters; (3) **Feature 3 (Decomposability):** Distance deterrence, origin production, and destination attraction are estimated independently; and (4) **Feature 4 (Practical Zero-Shot Setting):** The recovered deterrence parameters support survey-free gravity calibration in unseen cities without local target-city OD surveys.

4 Experimental Validation

We evaluate the proposed framework across 50 US metropolitan areas using LODES commuting origin–destination matrices, OpenStreetMap land-use features, and Meta Movement Distribution Maps (MDM) telemetry. The dataset is partitioned into 25 source training cities and 25 held-out target cities. Empirical evaluation proceeds systematically from recovery behavior to parameter consistency and downstream predictive performance.

First, evaluating recovery behavior, the probabilistic observation model (Eq. 4) establishes a multinomial log-likelihood over aggregate travel-distance bins. Across synthetic distributions and real Meta MDM telemetry, cross-entropy minimization yields practically useful recovery of Tanner parameters $\theta = (\alpha, \beta)$ without observing origin–destination pairs, providing empirical evidence of practical recoverability across the target metropolitan areas.

Next, evaluating parameter consistency, we compare parameters recovered from aggregate distance distributions (θ_{hist}) against reference parameters calibrated on complete LODES OD matrices (θ_{OD}) across 25 target cities. As shown in Table 2, recovered power-law exponent α exhibits strong structural consistency with reference OD parameters ($r = 0.909$, median relative error 1.7%, 100% of cities within ±10%). Exponential decay rate β shows a tight absolute margin of error (MAE = 0.0026). Figure 2 demonstrates that aggregate-identified deterrence curves closely track full OD calibration. Furthermore, in a separate case study using a coarse 3-bin summary of Meta MDM telemetry, models using Meta MDM identified parameters achieve a downstream Common Part of Commuters (CPC) of 0.7080 under oracle outflows (Table 3), showing that even coarse aggregate representations can remain empirically informative for parameter recovery within the proposed framework. This 3-bin case study uses a coarser aggregate resolution than the main cross-city benchmark reported below, so the two CPC figures are not directly comparable.

Finally, evaluating downstream predictive consequence, we assess flow predictions generated using aggregate-identified parameters against baseline models across 25 held-out target cities (Table 4). This benchmark uses the standard aggregate resolution adopted throughout the main evaluation, so its CPC values are not directly comparable to the single coarse 3-bin case study reported above. Under oracle outflow control, a Two One-Sided Test (TOST) (Lakens 2017) confirms that gravity models using Meta MDM identified deterrence parameters (CPC 0.7337) achieve statistical equivalence with models calibrated on

Algorithm 1: Probabilistic Parameter Identification and Gravity Calibration Framework

Input: Source cities spatial data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate distance distribution $\{b_k\}_{k=1}^{K}$
Output: Identified deterrence parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ and predicted target OD matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Parameter Identification

1. Compute geometric spatial exposure E_k for target city using spatial layout $\mathbf{X}^{(t)}$ via Eq. (3).
2. Identify latent Tanner parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ by minimizing cross-entropy $H(\mathbf{b}, \boldsymbol{\pi}(\theta))$ via Eq. (5).

Phase 2: Gravity Calibration & Downstream OD Generation

3. Estimate target city origin production: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
4. Compute target city destination attraction: $\hat{A}_j^{(t)} = \frac{1}{2}[\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})]$.
5. Generate downstream OD flows $\hat{T}_{ij}^{(t)}$ using Eq. (6) with identified parameters $(\hat{\alpha}, \hat{\beta})$.

Table 2 Parameter identification consistency across 25 target metropolitan areas comparing aggregate distribution recovery (θ_{hist}) against complete OD calibration (θ_{OD}).

Parameter	Pearson r	MAE	Median Rel. Error (%)	Cities within ±10%	Statistical Ev
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	High structural across observat
Exponential rate (β)	0.427	0.0026	0.0%	72.0%	Tight absolute gin between estimates.

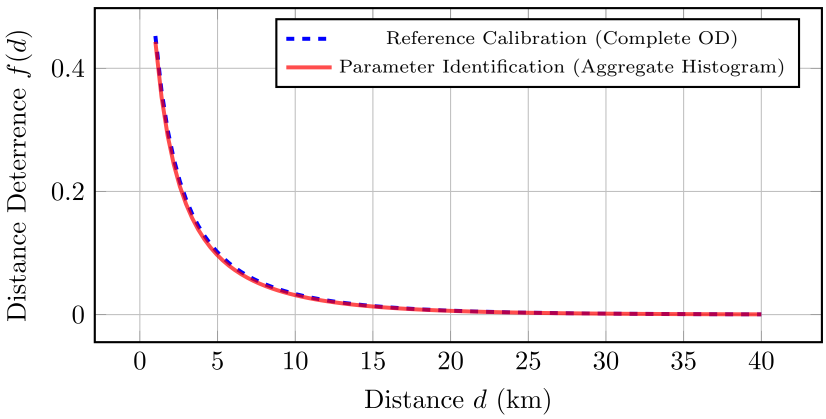

Fig. 2 Comparison of Tanner deterrence curves $f(d) = (d + \delta)^{-\alpha} \exp(-\beta d)$ parameterized by complete OD calibration (dashed blue) versus aggregate travel-distance distribution recovery (solid red).

complete local OD surveys (CPC 0.7382) within a pre-specified equivalence margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Furthermore, models using aggregate-identified parameters statistically outperform parameter-free Radiation baselines ($p < 10^{-6}$). Figure 4 illustrates predicted vs. observed flow agreement.

To validate component necessity, game-theoretic Shapley value analysis (Shapley 1953) across all $2^3 = 8$ coalitions (Table 5) reveals that relative to a uniform flow baseline (CPC 0.4263), the distance-deterrence component $f(d)$ accounts for a mean CPC gain of 0.2258 (**85.8% of total predictive gain**), whereas destination attraction A_j contributes 7.9% (0.0208) and origin production O_i contributes 6.4% (0.0167) (Figure 3). Parameter identification remains highly stable across heterogeneous urban morphologies, from

Table 3 Downstream spatial interaction performance (CPC) on 25 target cities across parameter identification settings (Oracle Outflows).

Identification Setting	**Data Source**	**Downstream CPC**	**CPC Gap vs. Reference**
Reference Calibration	Complete OD matrix	0.7411	Reference
20 equal-width distance bins	Aggregate GT Histogram	0.7403	-0.0008
3 physical distance bins	Aggregate GT Histogram	0.7348	-0.0063
3 physical distance bins	Meta Movement Dist. Maps	0.7080	-0.0331

monocentric cores (e.g., Boston) to polycentric sprawling regions (e.g., Atlanta).

5 Discussion

The principal empirical finding of this study is that, within the specified gravity model and aggregate observation framework, aggregate travel-distance distributions are empirically informative enough to support practical recovery of latent distance-deterrence parameters despite discarding origin–destination identities. Rather than demonstrating that aggregate data reconstruct point-to-point flows directly, this work shows that aggregate observations can remain empirically informative for parameter recovery. This suggests a useful reframing: survey-free mobility modeling can be viewed as a parameter-identification problem under limited observations rather than solely as a flow reconstruction task.

Methodologically and practically, the proposed framework separates parameter identification from downstream flow generation. Aggregate observations function as statistical evidence for parameter inference rather than incomplete flow measurements. The clear decomposition into recovering distance deterrence from aggregate distance observations while estimating origin production and destination attraction separately is well aligned with the survey-free zero-shot calibration objective. Practically, the framework supports survey-free gravity-model calibration for data-scarce urban regions lacking observed OD matrices when compatible aggregate distance summaries are available.

We delineate the scope and limitations of this investigation. Scientifically, the study focuses on identifying distance-deterrence parameters within gravity-model formulations; individual psychological travel choices remain outside the probabilistic framework. Methodologically, the formulation is established under singly constrained gravity assumptions. Experimentally, validation is conducted across 50 metropolitan areas within one national context, and the observation model considers one-dimensional aggregate travel-distance distributions. Formal questions of identifiability, information preservation, and statistical sufficiency remain open theoretical problems beyond the scope of the present empirical study.

Looking forward, promising research directions include formulating higher-dimensional aggregate mobility representations, exploring alternative distance-deterrence functional forms, and testing related observation models beyond the present gravity setting. More broadly, the results suggest that the usefulness of an observation for parameter recovery depends on the information it preserves, not only on whether it records complete origin–destination pairs. Future work should test how far this empirical conclusion extends across other observation types, model families, and geographic settings.

6 Conclusion

Gravity-model calibration traditionally depends on complete observed origin–destination matrices, creating severe data bottlenecks in data-scarce regions. Empirical evaluation across 50 US metropolitan areas shows that, within the specified gravity model and aggregate observation framework, aggregate travel-distance distributions are empirically informative enough to support practical recovery of city-specific distance-deterrence parameters without observing origin–destination interactions. In particular, the dominant power-law component is recovered with strong agreement to OD-based calibration, while downstream spatial interaction flows generated using the recovered parameters achieve statistical equivalence with local OD-calibrated models

Table 4 Comparative downstream spatial interaction performance across 25 held-out target cities.

Model Framework	Mean CPC	Std Dev	Type	Target OD Needed?
Group 1: Survey-Free Deployment (No target OD data)				
Proposed (Meta MDM Identified Deterrence)	0.6860	0.0391	Decomposed	No
Proposed (National Fallback Deterrence)	0.6896	0.0384	Decomposed	No
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	No
Group 2: Oracle Diagnostic (True Outflows $O_i^{\mathbf{true}}$)				
Proposed (Oracle O_i, Meta MDM Deterrence)	0.7337	0.0256	Decomposed	No (Deterrence only)
Proposed (Oracle O_i, National Fallback Deterrence)	0.7398	0.0248	Decomposed	No (Deterrence only)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	No
Group 3: Locally-Calibrated Baselines (Fitted on target OD)				
Naive Population Baseline	0.6758	0.0384	Baseline	No
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	Yes
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	Yes
Radiation Model	0.4851	0.0516	Param-free	No

Table 5 Factorial Ablation and Shapley Value Information Attribution (Average across $N = 25$ target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Predictive Cont
Distance-Deterrence Component $f(d)$	-34.9%	0.2258	85.8
Destination Attraction A_j	-4.7%	0.0208	7.9
Origin Production O_i	-3.6%	0.0167	6.4

Fig. 3 Shapley value attribution illustrating that distance-deterrence friction accounts for 85.8% of predictive gain in spatial interaction modeling.

within a 0.01 CPC margin, with distance friction accounting for 85.8% of total predictive gain.

Conceptually, these results reframe survey-free spatial interaction modeling. Rather than treating aggregate mobility products as incomplete substitutes for OD matrices, they can be regarded as observation layers whose preserved information may support parameter recovery under explicit modeling assumptions. This study provides an empirical starting point for testing which aggregate observations are informative enough to recover distance-dependent spatial interaction parameters in practice.

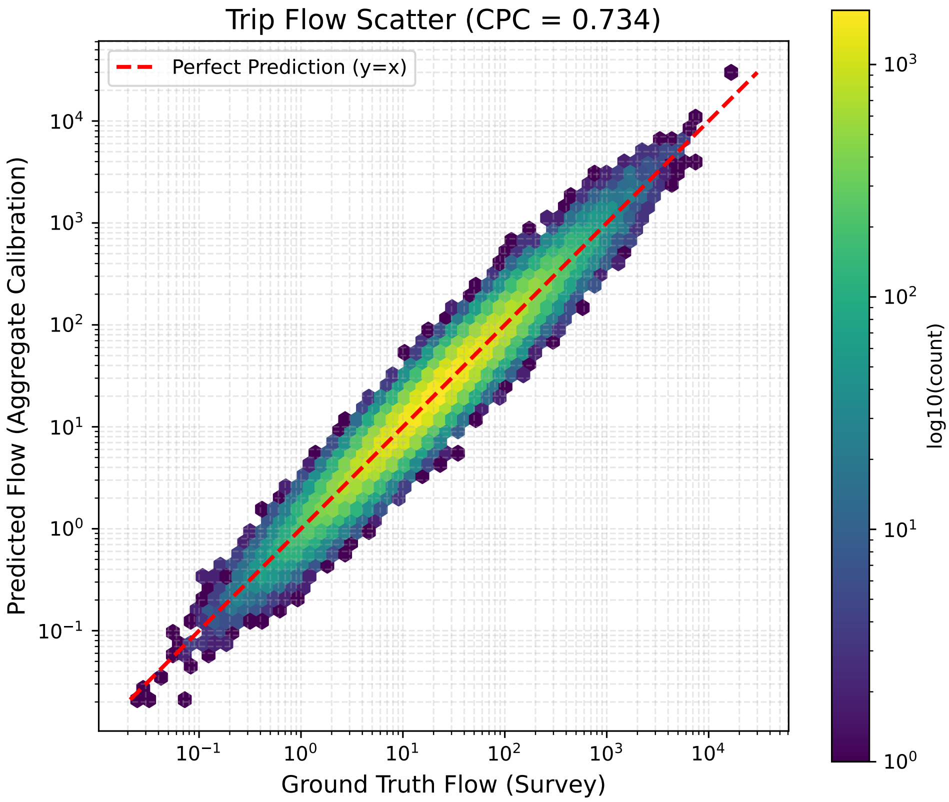

Fig. 4 Scatter plot comparing predicted flows against ground-truth observations across 25 target cities. Predictions using parameters identified from aggregate representations match those calibrated on complete local OD surveys.

Appendix A Empirical Recovery Evidence

A.1 Parameter Identification across Cities

Recovered parameter trajectories and optimization surfaces across target metropolitan areas.

A.2 City-wise Parameter Estimates

Detailed identification statistics for all 25 held-out target cities.

A.3 Representative Case Studies

Detailed analyses of representative metropolitan areas contrasting monocentric and polycentric urban morphologies.

A.4 Parameter Consistency

Statistical agreement measures between aggregate-identified parameters and complete OD-calibrated references.

A.5 Aggregate Representation Resolution

Sensitivity analysis evaluating parameter identification accuracy across histogram resolutions $K \in \{3, 5, 10, 20\}$.

A.6 Cross-City Identification Stability

Distribution of recovered Tanner parameters across all evaluation cities.

Appendix B Methodological Validation

B.1 Full Factorial Ablation

Tabular metrics for all $2^3 = 8$ component coalitions evaluated during Shapley value analysis.

B.2 Complete Shapley Analysis

Mathematical derivation and cooperative game formulation for predictive information attribution.

B.3 Feature Importance

SHAP feature rankings for origin-production estimation.

B.4 Feature Sensitivity

Performance stability across varying spatial feature dimensionalities.

B.5 Observation Exposure Correction

Mathematical demonstration of geometric spatial exposure correction E_k in preventing spatial layout bias.

B.6 Adaptive Self-loop Parameter

Validation of adaptive zonal offset $\delta = 0.5\bar{R}_{\text{zone}}$.

B.7 Attraction Modeling

Supplementary evaluation of destination attraction specifications.

Appendix C Reproducibility Details

C.1 Overall Pipeline

Overview of computational workflow from raw data ingestion to evaluation.

C.2 Datasets

Data schemas for LODES, OpenStreetMap, and Meta Movement Distribution Maps.

C.3 Hyperparameters

Hyperparameter specifications for machine learning components and L-BFGS-B optimization.

C.4 Optimization

Convergence criteria and loss trajectory formulations.

C.5 Runtime

Computational execution times per city and pipeline stage.

C.6 Hardware

Hardware environment details used for empirical experiments.

C.7 Random Seeds

Random seed specifications for experimental reproducibility.

C.8 Data Splits

List of 25 source training cities and 25 target held-out evaluation cities.

C.9 Code Availability

Open-source repository details and execution scripts.

C.10 Computational Complexity

Theoretical and empirical scaling complexity analysis.

Appendix D Mathematical Foundations

D.1 Observation Operator

Derivation of observation operator $\mathcal{P}$ mapping full spatial interaction space onto distance histograms.

D.2 Probabilistic Observation Model

Derivation of categorical and multinomial trip distributions over distance bins.

D.3 Likelihood Derivation

Step-by-step derivation of aggregate multinomial log-likelihood.

D.4 Cross-Entropy Equivalence

Formal proof that cross-entropy minimization is equivalent to maximum likelihood estimation under normalized distributions.

D.5 Observation Exposure Correction

Mathematical derivation of geometric exposure normalization E_k.

D.6 Empirical Recovery Evidence

Empirical and statistical evidence supporting Tanner deterrence identification.

D.7 Computational Complexity

Theoretical computational bounds for parameter identification algorithms.

Appendix E Extended Discussion

E.1 Why Tanner Deterrence?

Methodological justification for selecting the two-parameter Tanner deterrence function.

E.2 Why Aggregate Travel-Distance Distributions?

Information-theoretic rationale for choosing trip-length distributions as the statistical representation layer.

E.3 Observation versus Reconstruction

Conceptual distinction between parameter identification from aggregate representations and direct flow matrix reconstruction.

E.4 Observation-Centered Spatial Interaction Modeling

Perspective connecting probabilistic parameter identification with physics-informed AI and general urban interaction science.

References

Barbosa H, Barthelemy M, Ghoshal G, et al (2018) Human mobility: Models and applications. Physics Reports 734:1–74. https://doi.org/10.1016/j.physrep.2018.01.001

Bishop CM (2006) Pattern Recognition and Machine Learning. Springer, New York

Buckee CO, Balsari S, Chan J, et al (2020) Aggregated mobility data could help fight covid-19. Science 368(6487):145–146

Casella G, Berger RL (2002) Statistical Inference, 2nd edn. Duxbury Press, Pacific Grove, CA

Cover TM, Thomas JA (1991) Elements of Information Theory. John Wiley & Sons, New York, https://doi.org/10.1002/0471200611

Egu O, Bonnel P (2020) How comparable are origin–destination matrices estimated from automatic fare collection, origin–destination surveys and household travel survey? an empirical investigation in lyon. Transportation Research Part A: Policy and Practice 138:267–282. https://doi.org/10.1016/j.tra.2020.05.021

Enaya A, Zhong C, Batty M, et al (2026) TransGM: Transferable gravity models for cross-city policy transfer. Computers, Environment and Urban Systems https://doi.org/10.1016/j.compenvurbsys.2026.102455

Flowerdew R, Aitkin M (1982) A method of fitting the gravity model based on the poisson distribution. Journal of regional science 22(2):191–202

Fotheringham AS, O'Kelly ME (1989) Spatial interaction models: formulations and applications. Kluwer Academic Publishers

González MC, Hidalgo CA, Barabási AL (2008) Understanding individual human mobility patterns. Nature 453(7196):779–782. https://doi.org/10.1038/nature06958

Hussain E, Bhaskar A, Chung E (2021) Transit od matrix estimation using smartcard data: Recent developments and future research challenges. Transportation Research Part C: Emerging Technologies 125:103044. https://doi.org/10.1016/j.trc.2021.103044

Hyman GM (1969) The calibration of trip distribution models. Environment and Planning A 1(1):105–112. https://doi.org/10.1068/a010105

Lakens D (2017) Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. Social Psychological and Personality Science 8(4):355–362. https://doi.org/10.1177/1948550617697177

Lenormand M, Bassolas A, Ramasco J (2016) Systematic comparison of trip distribution laws and models. Journal of Transport Geography 51:158–169. https://doi.org/10.1016/j.jtrangeo.2016.02.003

Merlin LA (2020) A new method using medians to calibrate single-parameter spatial interaction models. Journal of Transport and Land Use 13(1):49–70. https://doi.org/10.5198/jtlu.2020.1614

Meta (2026) Movement distribution maps. https://ai.meta.com/ai-for-good/datasets/movement-distribution-maps/, aI for Good Datasets. Accessed: 27 June 2026. No DOI assigned.

de Montjoye YA, Hidalgo CA, Verleysen M, et al (2013) Unique in the crowd: The privacy bounds of human mobility. Scientific reports 3(1):1–5

Murphy KP (2012) Machine Learning: A Probabilistic Perspective. MIT Press, Cambridge, MA

Oliver N, Lepri B, Sterly H, et al (2020) Mobile phone data for informing public health actions across the covid-19 pandemic life cycle. Science advances 6(23):eabc0764

Pappalardo L, Manley E, Sekara V, et al (2023) Future directions in human mobility science. Nature Computational Science 3:588–600. https://doi.org/10.1038/s43588-023-00469-4

Sen A, Smith TE (1995) Gravity models of spatial interaction behavior. Springer

Shapley LS (1953) A value for n-person games. In: Contributions to the Theory of Games, vol 2. Princeton University Press, p 307–317, classic book chapter; no DOI assigned

Simini F, Gonzalez M, Maritan A, et al (2012) A universal model for mobility and migration patterns. Nature 484:96–100. https://doi.org/10.1038/nature10856

Simini F, Barlacchi G, Luca M, et al (2021) A deep gravity model for mobility flows generation. Nature Communications 12(1):5647. https:

//doi.org/10.1038/s41467-021-26752-6

Song C, Qu Z, Blumm N, et al (2010) Limits of predictability in human mobility. Science 327(5968):1018–1021. https://doi.org/10.1126/science.1177170

Tanner J (1961) Factors affecting the amount of travel. Road Research Technical Paper 51, Road Research Laboratory, London

Wilson A (1971) A family of spatial interaction models, and associated developments. Environment and Planning A 3(1):1–32. https://doi.org/10.1068/a030001

Yang J, Huang S, Huang Z, et al (2026) Transferable human mobility network reconstruction with neurogravity. Nature Computational Science 6:630–641. https://doi.org/10.1038/s43588-026-01003-y

Yang Y, Herrera C, Eagle N, et al (2014) Limits of predictability in commuting flows in the absence of data for calibration. Scientific Reports 4:5662. https://doi.org/10.1038/srep05662